THE HISTORY ARRIVES BEFORE THE PATIENT DOES.

ED Triage Pre-Arrival Intake Assistant

A caregiver's chat that starts after the decision to go to the emergency department, asks what a triage nurse would ask, and hands the nurse a one-page, page-cited summary.

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A patient is on the way. The person beside them knows the history.

Today

- The family member waits with nothing to do
- The nurse collects the history by voice at the counter
- Facts that change urgency may never be mentioned

With the assistant

- The chat asks the triage questions during the trip
- Each answer becomes a structured, sourced fact
- Urgency is checked against page-cited ESI rules as facts arrive
- The nurse scans a QR code and starts informed

One journey, four touchpoints

1

Decision to go

The chat starts only after this. It never advises whether to go.

2

During the trip

Danger signs first, then complaint, medications, history, infection screen.

3

At the door

The nurse scans a QR code — a plain link with a random session id.

4

At triage

One page: identity as declared, the urgency floor with its rule, the practice signal beside it, the AMPLE history with sources.

> No hospital procedure changes. The system adds information beside the existing process.

Nearby products, and why the ESI handbook

Five product classes sit nearby. None fills this slot.

- Telehealth consults (Doctor Anywhere): a pre-call form for one consultation
- Symptom checkers (Ada, Symptomate): advise whether to seek care — before the decision
- ED online check-in (NUHS app, InQuicker): queue and registration, not clinical history
- Pre-notification (Pulsara): paramedic to hospital, ambulance cases only
- In-ED patient apps (ERAdvisor): inform after arrival, collect nothing

Why ESI

- A fully published algorithm: four decision points (ENA, 2023)
- 150 worked cases in the fourth edition — an answer key by the authors (Gilboy et al., 2012)
- Inter-rater reliability 0.59 vs 0.49 for PACS on 300 Singapore patients (Fong et al., 2019)
- PACS publishes no criteria-level rulebook; a PACS mapping is an extension

Three technique groups, each built for real

Decision automation

MR

A page-cited ESI rulebook, interview-control rules and a deterministic engine that outputs a minimum urgency level.

Knowledge discovery

MR / RS

A classifier trained on 560,486 real ED visits, shown beside the rules; where rules and recorded practice diverge.

Cognitive techniques

CGS

A slot-filling interview whose next question is computed from the missing facts; schema-bound extraction; lay-to-clinical bridges.

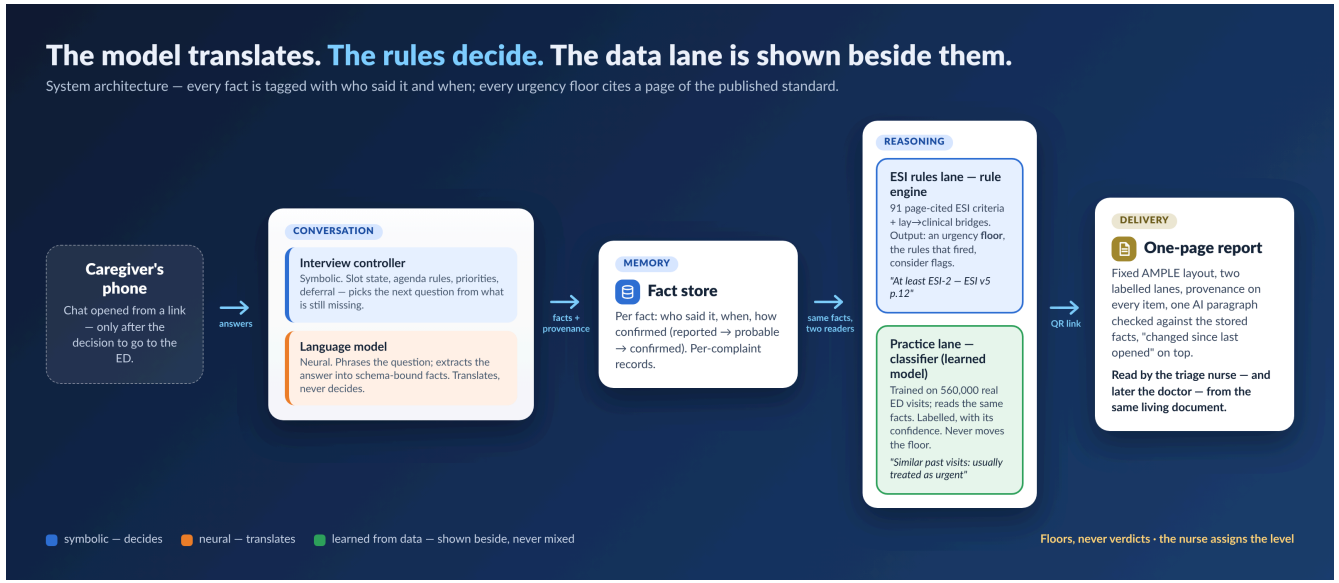
Out of scope: advice on whether to seek care · camera-based assessment · hospital IT integration · assigning the triage level — the nurse does that.

What the system learns from, and is measured against

DATASET	SIZE	ACCESS	USED FOR
Yale ED visits (Hong et al., 2018)	560,486 visits × 972 columns	public download	persona seeding · classifier training · ESI rules vs recorded practice
ESI practice cases (Gilboy et al., 2012)	150 worked cases	in the handbook	reference for encoding fidelity
Scripted personas	20–25	written by a teammate who does not build the chat	reference for interview accuracy
MIMIC-IV-ED demo (Johnson et al., 2023)	100 patients	open access	outcome checks

> Knowledge sources: ESI handbook 5th and 4th editions · CDC field-triage and NICE head-injury guidelines · AMPLE / SBAR / OLDCARTS. Every rule carries its source page and quoted span and is reviewed by a team member against that page.

The model translates. The rules decide.



Rule engine
chooses every question, evaluates answers against page-cited ESI criteria, outputs the floor: "at least ESI-2, pending vital signs"

Language model
phrases questions, turns answers into structured facts — decides nothing

Classifier
learned from real visits; shown beside the floor, never changes it. Two views, never mixed; the nurse assigns the level.

What the nurse sees



PRE-ARRIVAL INTAKE REPORT - session PANDA-8802 - version 2 - prototype output, advisory only

Patient AS DECLARED by caregiver
Tan Ah Kow DOB 3 March 1950

NOT VERIFIED
confirm against the patient

Answering-pattern note: 0 conflicting answer(s), 2 question(s) without a usable answer after three attempts (limb.screen, allergies_and_history). Answers of "don't know" are recorded as such and not counted here.

AT LEAST ESI-2

computed WITHOUT vitals - a floor, not a final level; nurse assessment prevails

B-30.anticoagulated-trauma
source: CDC Field Triage Guidelines (MMWR 2012 vol. 61 no. 1), Step Four special considerations, p.14 of the report (PDF p.16) · NICE NG232 (2023), recommendation 1.5.13 (PDF p.21)
"Anticoagulation and Bleeding Disorders: Patients with Head Injury Are at High Risk for Rapid Deterioration"

Pending: C (resources), D (vitals not available pre-arrival)

Practice signal (learned from past visits)
Most similar past visits (age 76, fall injury, other complaint, fall, anticoagulated) were ESI 1 or 2.
Does not change the floor above. About this signal

Consider flags (handbook: judgment, not automatic)

- B-27.geriatric-trauma-caution ESI v5 p.15 (Trauma, older adults)
- B-52.geriatric-vitals-medication-context-consider ESI v5 p.11 (Geriatric Considerations note)

Complaints (in order reported)

1. injury_fall presenting · reported - caregiver - 18:51

2. other reported - caregiver - 18:51

text	landed on right hip after slipping on wet floor	reported - caregiver - 18:51
onset	about an hour ago ~2026-09-26 17:51	reported - caregiver - 18:51
pain_rating	6	reported - caregiver - 18:51

Pain (0-10)

worst reported	**6**	reported - caregiver - 18:51

EVENTS & CONTEXT

mechanism	fall	reported - caregiver - 18:51

Identity

as declared by the caregiver — labelled not verified

Urgency floor

with the rule that produced it and its source

Practice signal

the classifier's one sentence, shown separately; never changes the floor

History with sources

every line tagged with who said it, when, and how certain; open questions stay visible

The prototype's report page for a fall on warfarin (session on the team demo, 26 Sep 2026).

Every claim gets a number

WHAT WE MEASURE	HOW	COMPARED AGAINST
Encoding fidelity	ESI v4 worked cases run through our rules; output vs the book's answer	the handbook's worked cases
Extraction accuracy	Facts stored vs the persona's script: precision and recall per category, at three levels of answer difficulty (direct, lay, vague)	scripted persona facts
Rules vs model-direct	Same cases, the model judging urgency directly: agreement, rerun variation, citation present	practice cases
End to end	A model-run interview that assigns a level on its own vs our rule-constrained interview; disagreements attributed	each persona's expected floor
Two-assessment agreement	Floor vs classifier on a held-out test set, with confidence intervals; classifier trained accuracy-optimised and class-balanced	recorded nurse levels
Floor soundness	Share of assessable visits where the floor exceeds the recorded nurse level — a bound on over-calling	recorded nurse levels

What could go wrong, and what we do about it

Verifying the rulebook

Every machine-drafted rule is checked by a person against its cited page — the largest block of manual work.

No clinical contact yet

Usefulness is assessed against published documents and data; an asynchronous review sheet is the fallback.

Lay words, clinical criteria

Mapping what a caregiver can say to what a rule reads is judgement work — each mapping cites a source.

Gaps in the source

Imprecise terms and inconsistent thresholds get expert review or conservative handling.

Honest evaluation

Scripted personas cannot reproduce everything real users do; some qualities will be reported as unvalidated.

In one line

Page-cited ESI rules decide. The model phrases and transcribes. A classifier is shown beside, never mixed.

> The history arrives before the patient does. — Questions